

Impact of the EU-ETS Indirect Cost Compensation

Evidence from French Manufacturing Plants

APE Master's Thesis

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Introduction and Motivation

EU-ETS: General Framework

- Cap-and-trade system, created in 2005, designed to price GHG emissions, notably covering the power and manufacturing sectors.
- Power producers pass carbon costs on to customers through higher prices.
- Risk of carbon leakage and reduced competitiveness for plants operating in power-intensive and internationally exposed sectors.

EU Indirect Cost Compensation: Covers French Plants Since 2015

- Offsets roughly 75% of indirect costs for plants operating in at-risk sectors.
- A group of non-subsidized plants also exists within the same sectors, since aid must be requested by the firm and is subject to several eligibility criteria.

Research Objectives

- Assess the impact of the compensation on beneficiary plants in terms of **electricity bills, consumption, and survival**, compared to non-beneficiaries.

Methodology Overview

Data and Construction of Plant-level Indicators

- List of beneficiaries and cumulative aid amounts (2015–2022) from the compensation registry for France, at the **firm-level**.
- Electricity records from the EACEI survey and activity status from the SIRENE database, at the **plant-level**.
- Treated plants identified using several firm-to-plant attribution methods.

Empirical Strategy

- Nearest-neighbor matching based on pre-treatment electricity consumption.
- Difference-in-differences and event studies using OLS TWFE, as well as the estimator by de Chaisemartin and d'Haultfœuille (2024).
- Extension of the analysis to heterogeneous treatment effects, depending on whether treated plants are covered by the EU ETS for their direct emissions.

Summary of Findings

Main Results

- **Positive effect on survival:** Treated plants show improved survival (20%), suggesting the policy helped mitigate rising energy costs.
- **Electricity bills:** Treated plants saw higher bill increases than controls, most likely due to a change in composition of the control group.
- **Heterogeneous effects:** EU-ETS treated plants faced increases in prices and bills compared to the overall control group, while non-EU-ETS treated plants experienced price reductions.

Comparison with Existing Literature

- **No rebound in consumption:** Unlike Basaglia et al. (2024), who find 22% increase in electricity use among treated, we find no significant consumption increase, possibly due to energy efficiency conditions in France (not in UK).
- They also find that output rose by 15–30%, carbon emissions increased by 22%, but there was no effect on energy efficiency. *Left for future analysis.*

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Policy Background

Eligibility Criteria: no minimum electricity consumption required.

- Eligible sectors: 14 sectors at risk of leakage (incl. aluminium, steel, paper).
- Applicants must provide electricity bill data verified by an accredited body.
- Large plants are also required to comply with **energy efficiency obligations**.

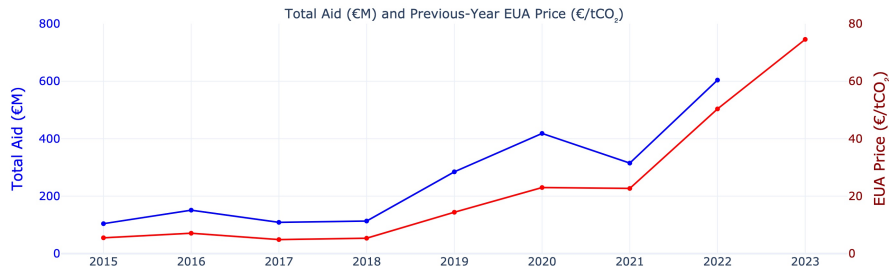
Aid Calculation: output-based subsidy, no electricity consumption involved.

- $\text{Aid} = 75\% \times \text{output} \times \text{emission factor} \times \text{efficiency benchmark} \times \text{CO}_2 \text{ price}$.
- Aid can reach 100% for heavily impacted firms (if the indirect costs remaining after base aid exceed 1.5% of the company's value added).

Funding, Amounts, and Beneficiaries

- Funded by revenues from EU ETS allowance auctions (max 25%).
- In France (2015–2022): around **2 billion** allocated; fewer than **200 plants** subsidized annually.

Compensation Aid Overview



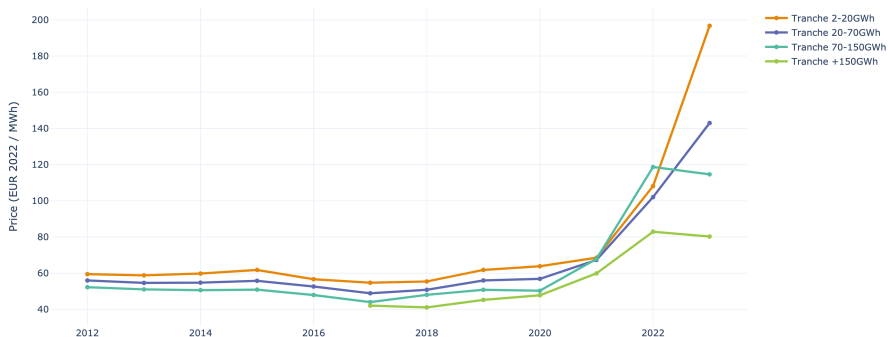
Evolution of Compensation Aid in France and EUA Prices (EU Commission).

Sector	NACE Code	Total Aid (M€)
Iron and Steel	C2410	471.9
Aluminium Metallurgy	C2442	440.1
Paper and Cardboard Manufacturing	C1712	250.9
Other Inorganic Basic Chemicals	C2013	245.8
Petroleum Refining	C1920	59.2

Top 5 sectors by total aid received (2015–2022).

Recent Electricity Price Trends

Electricity Prices for Industrial Consumers (excl. taxes & levies)



Electricity Prices in France by Consumption Tranche (Eurostat).

- Large consumers faced sharper increases due to market exposure and contract timing, unlike smaller ones with more stable, long-term contracts.
- Stresses the need to match treated and controls on consumption.

Identification of Treated Plants

Step 1: Assigning Treatment to Plants - “Sure Cases”

- Cross EACEI (plant-level) with the compensation registry (firm-level)
- Identify “sure cases”: firm has a **single plant** in a region receiving aid
- Use SIRENE db to verify no other plants exist under the same SIREN-Region
- For these 86 sure cases (35% of aid), assign full amount to the unique plant

Step 2: Ambiguous Cases (Multiple Plants) - “Self Method”

- Estimate regression: $\text{Aid} = f(\text{electricity bill})$ using sure cases (we find 13%)
- Predict aid for each plant within ambiguous SIREN-Region (multiple plants)
- Compare predicted aid to observed aid in registry; assign to the best match

Covers **95%** of total aid (86 sure + 55 ambiguous cases = 141 plants)

Construction of Final Datasets

Step 3: Equal Division Method - "AllDivMethod"

- Divide aid equally across all plants in a given SIREN-Region.
- Results in 186 treated plants, with smaller aid amounts per plant on average.

Construction of Final Panel Datasets (2008–2022) for Each Method

- Combine beneficiaries with all **non-beneficiary** SIRETs in the same sectors.
- Keep only plants observed **before and after 2015** (treatment year).
- Final datasets contain electricity prices, consumption, bills, and sector info.

Adding Survival Information: Yearly Activity Status .

- Combine plant's activity status in 2023 from the SIRENE db with the first year of only missing observations in EACEI to compute the shutdown date.
- Activity = 1 from the plant's creation until shutdown (no temporary halts).
- For robustness, an alternative index treats SIRETs with ID changes as active.

Summary Statistics on Raw Samples

- Number of treated: 68 (sure cases), 111 (self-method), 143 (allDivMethod).
- Number of controls: approximately 1,000 across all methods.

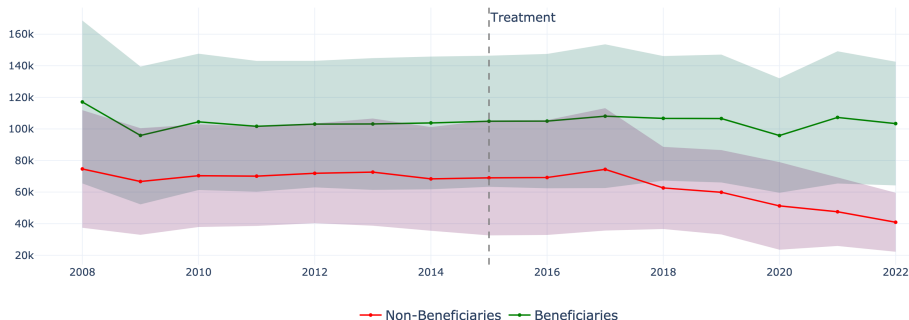
Variable	Treated	Controls
Price (€/MWh)	66.6	92.3
Quantity (GWh)	200.4	21.0
Bill (M€)	10.7	1.4
Total aid (M€)	1130	NA

*Summary statistics at the plant-level – self-method (unmatched).
Price, quantity, and bill averaged over the whole 2008–2022 period.*

- Large consumption disparity between treated and controls.
- Stress the need for matching to ensure valid comparisons.

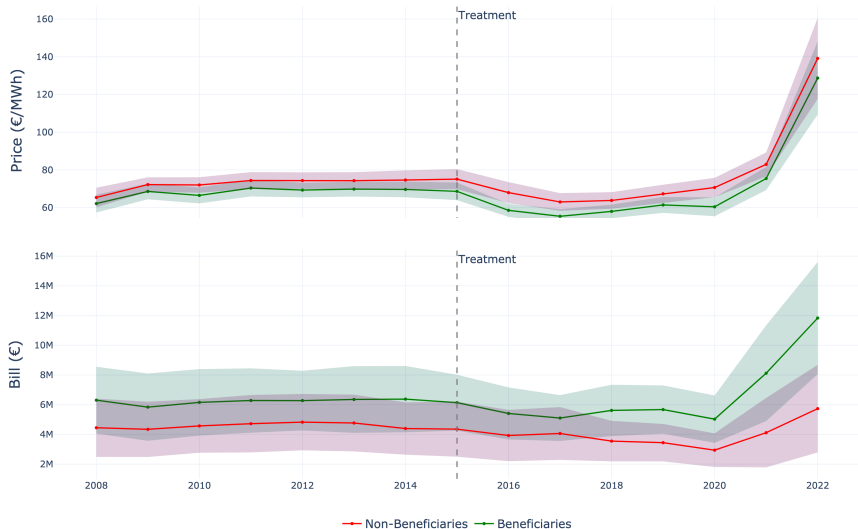
Matching Treated and Controls

- Nearest-neighbor algorithm with caliper (15%) and replacement.
- Based on mean electricity quantity in pre-period (2008–2014).
- Matches restricted within same 2-digit sector, can be one-to-many.



*Evolution of average quantity - self-method (matched).
90 beneficiaries, 313 non-beneficiaries (140 distinct).*

Matched Sample Characteristics



Price & Bill - self-method. Fewer controls over time.

Aid-to-Bill Ratios Across Samples

- Beneficiaries receive between 10.6% and 12.5% of their bill in compensation.
- Aligns with the 75% compensation rule assuming 14.4% of indirect costs.

Method	Unmatched	Matched
<i>Sure cases</i>	0.109 (0.07)	0.106 (0.07)
<i>Self-method</i>	0.125 (0.08)	0.116 (0.08)
<i>AllDivMethod</i>	0.190 (0.44)	0.200 (0.48)

Aid-to-bill = aid / electricity bill at the plant level. SE in parentheses.

- AllDivMethod** produces overly high ratios, questioning its accuracy.
- Self-method** ratios are closer to theoretical expectations.

Econometric Framework

Difference-in-Differences and Event-Study Strategy

- Use matched panel datasets (2008–2022) to compare treated and controls.
- Estimate treatment effects on bills, prices, consumption and survival.
- Control for plant and year FEs – and sector-by-year (2-digit) for robustness.

Main Specification (TWFE):

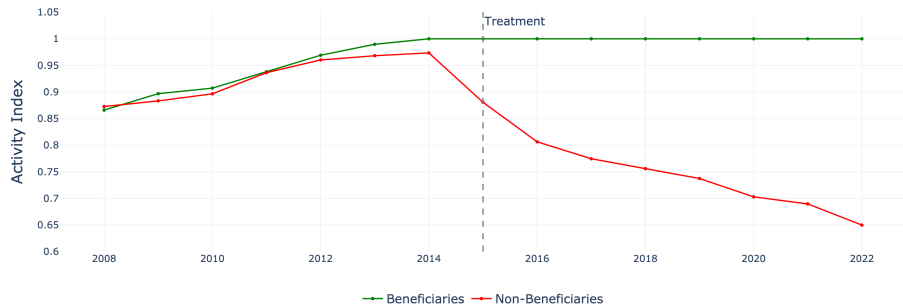
$$y_{it} = \alpha_i + \lambda_t + \beta \cdot \text{Treated}_i \times \text{Post}_t + \varepsilon_{it}$$

- α_i : plant fixed effects, λ_t : year fixed effects, β : average treatment effect.
- Annual interactions in event studies (baseline = 2014).
- Standard errors clustered at the 4-digit sectoral level.

Alternative Estimator: de Chaisemartin and d'Haultfœuille (2024)

- TWFE may yield biased estimates in incomplete panels by implicitly using already-treated units as controls. Can lead to negative weights, especially problematic under treatment effect heterogeneity.

Main Results on Survival Dynamics

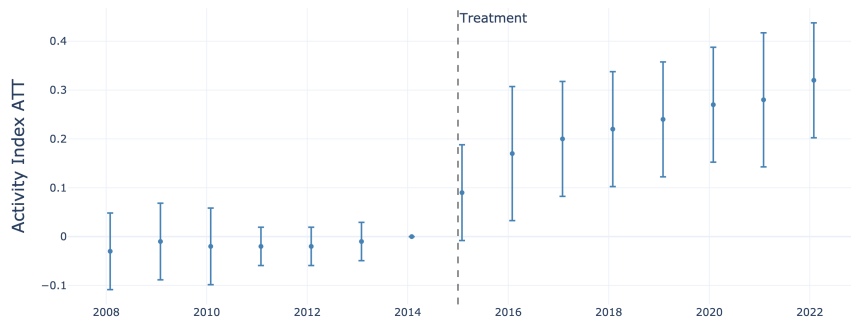


Mean activity index - matched without pre/post restriction (self-method).

Sample - self-method without pre/post restriction	Coef.	SE	R^2	N
unmatched	0.11***	(0.04)	0.45	27000
matched	0.24***	(0.07)	0.43	7110
robust matched (without SIRETs with ID changes)	0.05*	(0.03)	0.37	7110

Diff-in-diff results (OLS TWFE) for activity index (self-method)

Main Results on Survival Dynamics

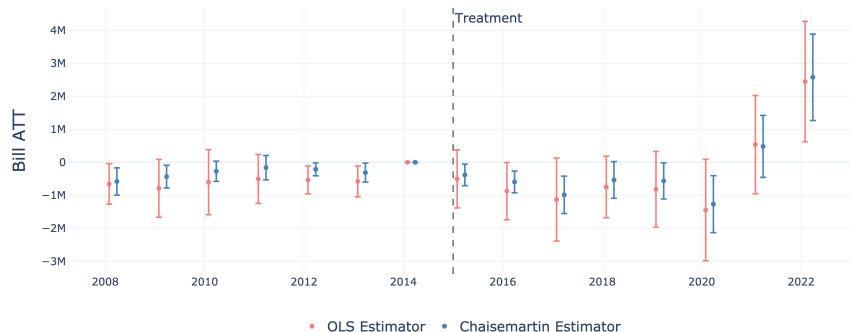


Event-study results (OLS TWFE) for activity index - matched (self-method)

Sample - self-method	Activity	Mean Quantity	Nb. of plants
<i>matched (without pre/post restriction)</i>	Closed	126.1 GWh/y	130
	Active	61.8 GWh/y	245

Electricity consumption in pre-period of exiting vs remaining **non-beneficiaries**

Main Results on Electricity-Related Outcomes

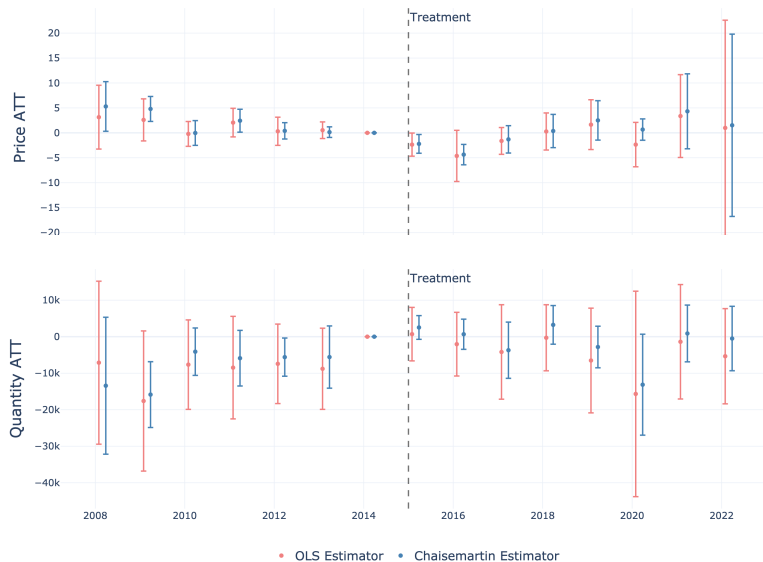


Sample	Outcome	OLS	Chaisemartin	R2
Self-Method (<i>N</i> = 4780)	Bill (k€)	130 (470)	-170 (190)	0.88
	Price (€/MWh)	-2.0 (1.4)	-0.5 (1.7)	0.61
	Quantity (MWh)	-2 220 (7 000)	-4 030 (2 900)	0.95
Sure Cases (<i>N</i> = 3140)	Bill (k€)	540* (300)	150 (110)	0.86
	Price (€/MWh)	-1.9 (1.2)	0.1 (1.5)	0.59
	Quantity (MWh)	3 840 (5 600)	-1 410 (2 700)	0.93

Electricity Price and Consumption: Self-Method



Electricity Price and Consumption: Sure Cases



EU-ETS Heterogeneity Analysis

Question: Does the treatment yield heterogeneous effects between EU ETS and non-EU ETS plants?

We estimate the following OLS regression:

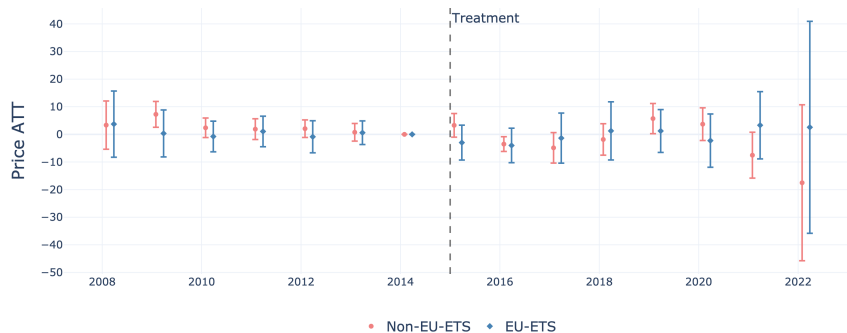
$$Y_{it} = \alpha_i + \lambda_t + \beta_1 \text{Treat}_{it} + \beta_2 (\text{Treat}_{it} \times \text{EUETS}_i) + \varepsilon_{it}$$

Interpretation:

- β_1 : ATT for non-EU-ETS treated plants.
- β_2 : additional effect for EU-ETS treated plants.
- $\beta_1 + \beta_2$: ATT for EU-ETS treated plants.

Standard errors clustered at the 4-digit sector level.

EU-ETS Heterogeneity Analysis - Price

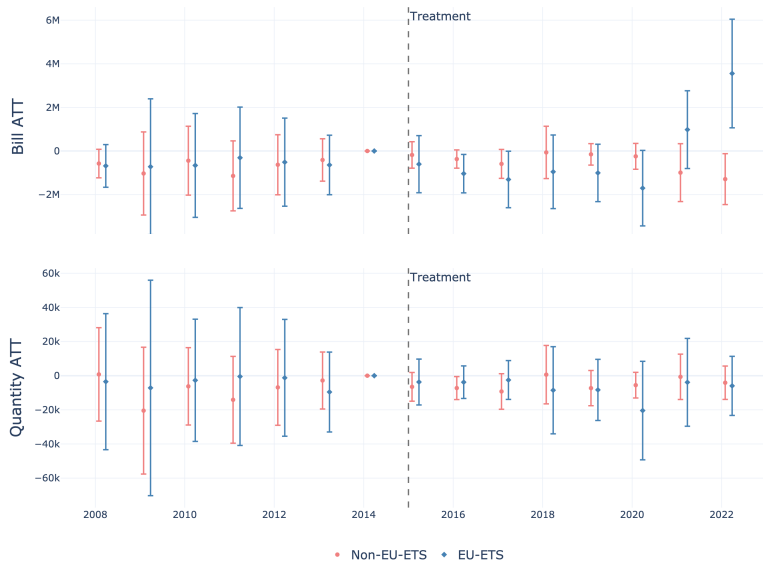


Event-study and Diff-in-diff results (OLS TWFE) for price

Method	Specification	β_1	β_2	R^2
Self-Method ($N = 4780$)	Standard TWFE	-5.0** (1.8)	4.3*** (1.4)	0.60
	Sector-year FEs	-5.2** (2.2)	4.3* (2.1)	0.61
Sure Cases ($N = 3140$)	Standard TWFE	-5.8** (2.2)	5.4** (2.1)	0.58
	Sector-year FEs	-5.4** (1.9)	4.6** (2.0)	0.59

Top 5 sectors by total aid received (2015–2022).

EU-ETS Heterogeneity Analysis - Bill and Consumption



Conclusion and Perspectives

Main Results

- **Positive effect on survival:** Treated plants show improved survival (20%).
- **Electricity bills:** Treated plants saw higher bill increases than controls, most likely due to a change in composition of the control group.
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Conclusion and possible extensions

- Would be valuable to further examine fuel switching and self-generation strategies adopted by non-beneficiaries.
- Future research could also investigate long-term investment behaviours, such as improvements in energy efficiency, alongside evolution of output, emissions, and broader impacts on productivity, employment, innovation.
- France committed to providing a total of approximately €13.5 billion in compensation between 2021 and 2030.

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